

Decent Coaching Classes

X SB (Semi English Medium)

Worksheet

Maths Part 1 : Chapters 1 to 3 | Maharashtra State Board (Semi English)

Show all steps of your working.

Chapter 1 : Linear Equations in Two Variables

Q1. Solve the following simultaneous equations using Cramer's rule.

$$3x - 4y = 10 ; 4x + 3y = 5$$

Q2. Solve the following word problem.

The sum of two numbers is 63 and their difference is 17. Find the numbers.

Chapter 2 : Quadratic Equations

Q3. Solve the following quadratic equation by factorisation.

$$2x^2 + 5x - 12 = 0$$

Q4. Solve the following word problem.

The product of two consecutive positive integers is 132. Find the integers.

Chapter 3 : Arithmetic Progression

Q5. Solve.

In an A.P., $t_4 = 22$ and $t_9 = 47$. Find the first term (a), the common difference (d) and t_{15} .

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Worksheet – Answer Key

Q1. $D = 3(3) - (-4)(4) = 25$; $Dx = 10(3) - (-4)(5) = 50$; $Dy = 3(5) - 10(4) = -25$. $x = Dx/D = 2$, $y = Dy/D = -1$.

Q2. Let the numbers be x and y . $x + y = 63$, $x - y = 17$. Adding: $2x = 80$, $x = 40$. Then $y = 23$. The numbers are 40 and 23.

Q3. $2x^2 + 5x - 12 = 0 \rightarrow 2x^2 + 8x - 3x - 12 = 0 \rightarrow 2x(x + 4) - 3(x + 4) = 0 \rightarrow (x + 4)(2x - 3) = 0$.
 $x = -4$ or $x = 3/2$.

Q4. Let the integers be x and $x + 1$. $x(x + 1) = 132 \rightarrow x^2 + x - 132 = 0 \rightarrow (x - 11)(x + 12) = 0$. $x = 11$ ($x = -12$ rejected as the integers are positive). The integers are 11 and 12.

Q5. $t_9 - t_4 = 5d \rightarrow 47 - 22 = 5d \rightarrow d = 5$. $t_4 = a + 3d = 22 \rightarrow a + 15 = 22 \rightarrow a = 7$. $t_{15} = a + 14d = 7 + 70 = 77$.